

Homework 4

Problem 1

求 $(2+x)^{\frac{3}{2}}/(1-x)$ 中 x^3 的系数

Answer:

$$\begin{aligned}\frac{(2+x)^{\frac{3}{2}}}{1-x} &= (2+x)^{\frac{3}{2}}(1+x+x^2+\cdots) \\ &= \sum_{k=0}^{\infty} \binom{\frac{3}{2}}{k} x^k 2^{\frac{3}{2}-k} (1+x+x^2+\cdots)\end{aligned}$$

其中 x^3 的系数为

$$\begin{aligned}&\binom{\frac{3}{2}}{0} 2^{\frac{3}{2}} + \binom{\frac{3}{2}}{1} 2^{\frac{1}{2}} + \binom{\frac{3}{2}}{2} 2^{-\frac{1}{2}} \\ &= 2^{\frac{3}{2}} + \frac{3}{2} 2^{\frac{1}{2}} + \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 2^{-\frac{1}{2}} \\ &= \frac{59}{16} \sqrt{2}\end{aligned}$$

Problem 2

求下列序列的生成函数（不能用无穷级数表示）

(1) $0, 0, 0, 0, -6, 6, -6, 6, \dots$

(2) $1, 0, 1, 0, 1, 0, \dots$

(3) $1, 2, 1, 4, 1, 8, \dots$

Answer:

(1)

$$\begin{aligned}(1, 1, 1, 1, \dots) &\Leftrightarrow \frac{1}{1-x} \\ (1, -1, 1, -1, \dots) &\Leftrightarrow \frac{1}{1-(-1)x} = \frac{1}{1+x} \\ (1, -1, 1, -1, \dots) &\Leftrightarrow -\frac{1}{1+x} \\ (6, -6, 6, -6, \dots) &\Leftrightarrow -\frac{6}{1+x} \\ (0, 0, 0, 0, 6, -6, 6, -6, \dots) &\Leftrightarrow -\frac{6x^4}{1+x}\end{aligned}$$

(2)

$$\begin{aligned}(1, 1, 1, 1, \dots) &\Leftrightarrow \frac{1}{1-x} \\ (1, -1, 1, -1, \dots) &\Leftrightarrow \frac{1}{1-(-1)x} = \frac{1}{1+x} \\ (2, 0, 2, 0, \dots) &\Leftrightarrow \frac{1}{1-x} + \frac{1}{1+x} \\ (1, 0, 1, 0, \dots) &\Leftrightarrow \frac{1}{2} \left(\frac{1}{1-x} + \frac{1}{1+x} \right)\end{aligned}$$

(3)

$$\begin{aligned}(1, 0, 1, 0, \dots) &\Leftrightarrow \frac{1}{2} \left(\frac{1}{1-x} + \frac{1}{1+x} \right) \\(0, 1, 0, 1, \dots) &\Leftrightarrow \frac{x}{2} \left(\frac{1}{1-x} + \frac{1}{1+x} \right) \\(0, 2, 0, 4, \dots) &\Leftrightarrow (0(\sqrt{2})^1, (\sqrt{2})^2, 0(\sqrt{2})^3, (\sqrt{2})^4, \dots) \\&\Leftrightarrow \sqrt{2} \cdot \frac{\sqrt{2}x}{2} \left(\frac{1}{1-\sqrt{2}x} + \frac{1}{1+\sqrt{2}x} \right) = x \left(\frac{1}{1-\sqrt{2}x} + \frac{1}{1+\sqrt{2}x} \right) \\(1, 2, 1, 4, 1, 8, \dots) &\Leftrightarrow \frac{1}{2} \left(\frac{1}{1-x} + \frac{1}{1+x} \right) + x \left(\frac{1}{1-\sqrt{2}x} + \frac{1}{1+\sqrt{2}x} \right)\end{aligned}$$

Problem 3

求 $i + 3j + 3k = n$, $i \geq 0, j \geq 1, k \geq 1$ 的整数解 $\langle i, j, k \rangle$ 的个数

Answer:

即求 $(1 + x + x^2 + \dots)(x^3 + x^6 + x^9 + \dots)(x^3 + x^6 + x^9 + \dots)$ 中 x^n 的系数

$$\begin{aligned}&(1 + x + x^2 + \dots)(x^3 + x^6 + x^9 + \dots)(x^3 + x^6 + x^9 + \dots) \\&= \frac{1}{1-x} \cdot \frac{x^3}{1-x^3} \cdot \frac{x^3}{1-x^3} = x^6(1+x+x^2)(1-x^3)^{-3}\end{aligned}$$

所以 x^n 的系数为

$$\binom{-3}{\frac{n-6}{3}}, \binom{-3}{\frac{n-7}{3}}, \binom{-3}{\frac{n-8}{3}}$$

取 $\frac{n-6}{3}, \frac{n-7}{3}, \frac{n-8}{3}$ 为整数的一项 (易知有且只有一项为整数)

Problem 4

$a(x)$ 为 $(a_0, a_1, a_2, \dots)$ 的生成函数, 求 $(a_0, a_0 + a_1, a_0 + a_1 + a_2, \dots)$ 的生成函数

Answer:

$$\begin{aligned}&a_0 + (a_0 + a_1)x + (a_0 + a_1 + a_2)x^2 + \dots \\&= a_0(1 + x + x^2 + \dots) \\&\quad + a_1(x + x^2 + x^3 + \dots) \\&\quad + a_2(x^2 + x_3 + x_4 + \dots) \\&\quad + \dots \\&= \frac{a_0}{1-x} + \frac{a_1x}{1-x} + \frac{a_2x^2}{1-x} + \dots \\&= \frac{1}{1-x}a(x)\end{aligned}$$